

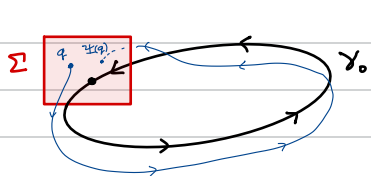
Hamiltonian Bifurcations

Setting 4-dim symplectic manifold (W, ω) , 1-param family of Hamiltonians $H_\mu: W \rightarrow \mathbb{R}$,
Periodic orbit γ_0 at $\mu=0$.

Question Change of orbit structure with varying μ . (= Bifurcation)

① Poincaré section and Floquet multiplier.

We can find a Poincaré section Σ at a point $p \in \gamma_0$,
which is a codim 1 submanifold transversal to γ_0 .



Define return map on Σ by

$$\mathcal{F}(q) = \text{Fl}_{\tau(q)}^{X_H}(q), \quad \tau(q) = \min \{t > 0 : \text{Fl}_t^{X_H}(q) \in \Sigma\}.$$

Note. $\mathcal{F}$ is not globally defined on Σ .

• $\Sigma \cap H^{-1}(c)$ is P.S. of $X_H|_{H^{-1}(c)}$ on $H^{-1}(c)$.

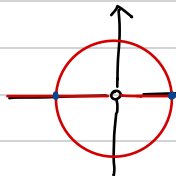
Floquet multiplier is an eigenvalue of linearized return map $d\mathcal{F}$, say λ .
If $\lambda \neq 1$, γ_0 is called non-degenerate.

$d\mathcal{F} \in \text{Sp}(2)$, so eigenvalues are in pair $\{\lambda, \bar{\lambda}\}$.

But it's 2-dimensional, which means $\lambda = \bar{\lambda}$ or $\lambda^{-1} = \bar{\lambda}$.

• If $\lambda = \bar{\lambda}$, so $\lambda \in \mathbb{R}$, we call γ_0 is hyperbolic.

• If $\lambda^{-1} = \bar{\lambda}$, so $\lambda \in S^1$, we call γ_0 is elliptic.



As μ varies, λ also varies continuously.

In our 2-dim case, the transition between hyp. & ell.
occurs only if $\lambda = \pm 1$.

Prop If γ_0 is non-degenerate, there exists an orbit cylinder along μ ,
i.e. $\exists \Gamma: S^1 \times (-\varepsilon, \varepsilon) \xrightarrow{C^\infty} W$ s.t. $\Gamma(t, \mu) := \gamma_\mu(t)$ is a periodic orbit of H_μ .

"The bifurcation is a breaking of an orbit cylinder."

② Birkhoff normal form.

To study bifurcation, we study the periodic points of return map $\mathbb{Z}_k$.

Meyer 70, used Birkhoff normal form to study this.

We introduce somewhat simplified version.

If $\lambda=1$, a bif. happens at γ_0 .

If $\lambda^k=1$, a bif. happens at γ_0^k , k -th cover of the orbit.

Simplifying steps.

- i) $\mathbb{Z}$ can be written as time 1-flow of time-dependent Ham. G_t .
 - ii) Quadratic part of G_t can be chosen time-independent.
 - iii) (Birkhoff normal form) $\exists$ Formal power series $G(t) = \sum_{k \geq 2} G_k(t)$, $\deg G_k = k$,
s.t. $\{G_2^{ss}, G_k\} = \dot{G}_k$, where G_2^{ss} is semisimple part (induces a vector field).
- If $\lambda=1$, $G_2 = \frac{\alpha}{2} V^2$ (generates $\begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix}$) so $G_2^{ss} = 0$.
 $\Rightarrow G$ can be taken autonomously.
 - If $\lambda = e^{2\pi i l/k}$, $G_2 = \frac{\pi l}{k} (u^2 + v^2) = G_2^{ss}$.
However, if we apply this procedure for $\mathbb{Z}_k$, we get $G_2 = G_2^{ss} = 0$.
Meanwhile, we want G to be equivariant to $\mathbb{Z}_k$ -action by $e^{2\pi i l/k}$.
 $\Rightarrow G$, which generates $\mathbb{Z}_k$, can be taken autonomous $\mathbb{Z}_k$ -polynomial.

Now we write $P_k, P_k^\circ \in \mathbb{R} \llbracket u, v \rrbracket_{\mathbb{Z}_k}$, where the quadratic part of P_k° is nilpotent; i.e. $P_{k,2}^\circ = \frac{\alpha}{2} V^2$ or 0.

Prop If $\lambda^k=1$, then critical points of $G_\mu = P_k^\circ + \mu P_k$ corresponds to the periodic orbit appears in a bifurcation.

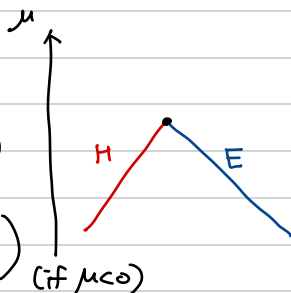
③ Meyer's classification.

i) $k=1$. (Birth-death)

$$G_\mu = \frac{\alpha}{2} v^2 + \frac{\beta}{3} u^3 + \mu(au + bv). \quad (\alpha, \beta, a, b > 0)$$

$$\Rightarrow \nabla G_\mu = \begin{pmatrix} \beta u^2 + a\mu \\ \alpha v + b \end{pmatrix} = 0, \text{ so } \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \pm \sqrt{-a\mu/\beta} \\ -b/\alpha \end{pmatrix} \text{ (if } \mu < 0)$$

i.e. $\exists 2$ orbits if $\mu < 0$, 0 orbits if $\mu > 0$.

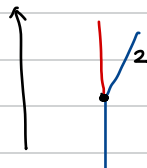


Note By computing Hessian, we can check the stability (elliptic/hyperbolic).

ii) $k=2$. (Period doubling)

$$\mathbb{R}[u, v]_{\mathbb{Z}_2} = \mathbb{R}[u^2, uv, v^2].$$

$$\Rightarrow G_\mu = \frac{\alpha}{2} v^2 + \frac{\beta}{4} u^4 + \mu \cdot \frac{a}{2} u^2 \quad (\alpha, \beta, a > 0)$$



$$\nabla G_\mu = \begin{pmatrix} \beta u^3 + \mu \cdot a u \\ \alpha v \end{pmatrix} = 0, \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \pm \sqrt{-a\mu/\beta} \\ 0 \end{pmatrix} \text{ (if } \mu < 0)$$

$\Rightarrow \mu > 0$: 1 orbit (cor. to 0)

$\mu < 0$: 1 orbit (, ,)

2 periodic points of $\mathbb{Z}_2$, which forms an orbit of period 2.

Note As μ passes 0, the stability type changes.

iii) $k=3,4$ (phantom kirs)

$$|R[u,v]|_{\mathbb{Z}_k} = |R[\mathbb{I}z^2, \bar{z}^k, \bar{z}^k]|_{\mathbb{Z}_k} = |R[r^2, r^k \cos k\theta, r^k \sin k\theta]|.$$

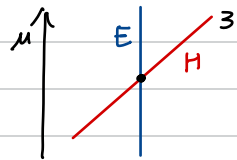
$$G_\mu = \frac{\alpha}{3} r^3 \cos 3\theta + \mu \cdot \frac{a}{2} r^2$$

$$\nabla G_\mu = \begin{pmatrix} \alpha r^2 \cos 3\theta + \mu \cdot ar \\ -\alpha r^3 \sin 3\theta \end{pmatrix} = 0 \Rightarrow r=0 \text{ \& } \sin 3\theta = 0 \Rightarrow \cos 3\theta = \pm 1$$

$$r = -\mu a / \alpha \cos 3\theta$$

i.e. $\mu < 0$: $r=0$ & 3 points ($\cos 3\theta = 1$) $= \pm \mu a / \alpha$

$\mu > 0$: $r=0$ & 3 points ($\cos 3\theta = -1$)



iv) $k \geq 4$ (Emission)

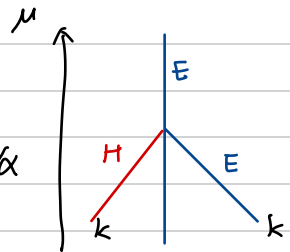
$$G_\mu = \frac{\alpha}{4} r^4 + \frac{\beta}{k} r^k \cos k\theta + \mu \cdot \frac{a}{2} r^2$$

$$\nabla G_\mu = \begin{pmatrix} \alpha r^3 + \mu \cdot ar \\ -\beta r^k \sin k\theta \end{pmatrix} = 0 \Rightarrow r=0, \sin k\theta = 0$$

$$r^2 = -\mu a / \alpha$$

i.e. $\mu < 0$: $r=0$ & $2k$ -points

$\mu > 0$: $r=0$



The borderline case $k=4$, as have both iii) & iv) case, depending on the coefficient of r^4 & $r^4 \cos 4\theta$.